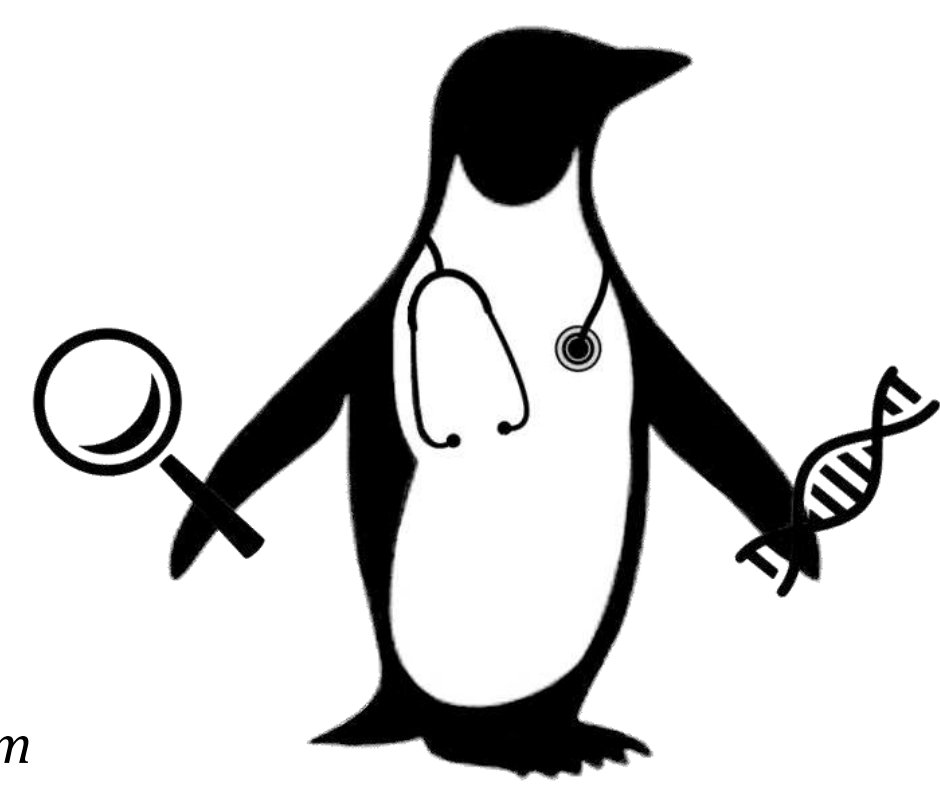


PENGQUIN: PErsoNal Genome QUery IN Healthcare and Clinical Practice



Elias Crum^{1,2}

¹IDLab, Department of Electronics and Information Systems, Ghent University – imec, Belgium
²Flemish institute for Technological Research (VITO) Mol, Belgium

Genome Data for Clinical Care?

Still Many challenges:

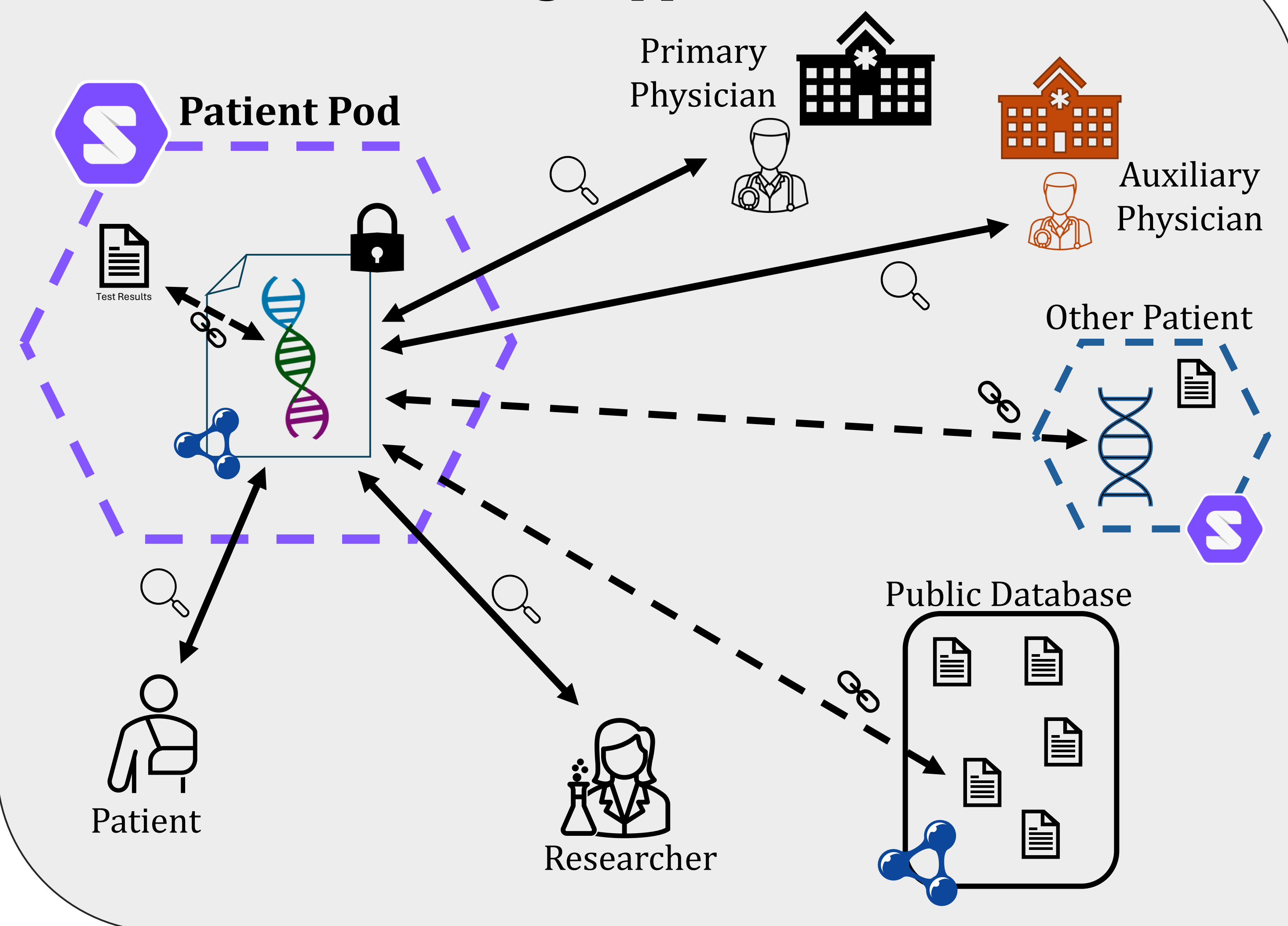
- Storage
- Privacy
- Governance
- Formatting

Could Semantic Web technology help?

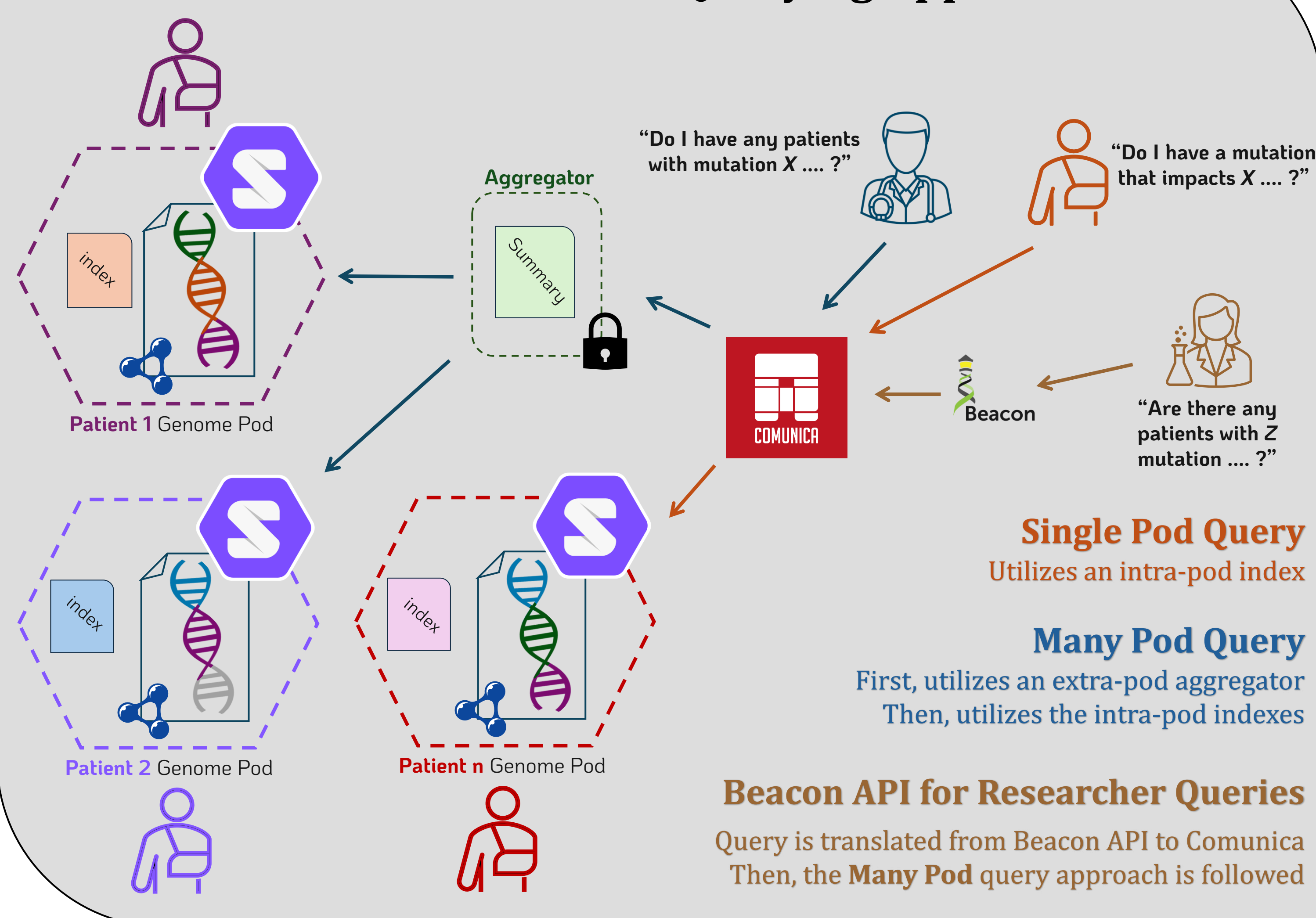
Personal Genome Sequences

- Variant Call Format (VCF)
- Very large (~GBs)
- Strict privacy protections

A Patient-Centric Storage Approach



An Index-Based Link Traversal Querying Approach



Research Challenges

- Privacy granularity
 - ↳ Index privacy concerns?
 - ↳ How granular is adequate?
- RDF Conversion + Data Linkages
 - ↳ VCF to RDF Conversion?
 - ↳ What data should be linked?
 - ↳ Automation of data linking?
- Querying performance
 - ↳ Many, large data sources
 - ↳ Algorithms that use indexes?

Engineering Tasks

- Genomic data storage in Solid pods
- Data privacy management
- Data Sharing via the web

Research Pursuits

- Represent genomic data as RDF?
- Linking genomic and health data?
- Single and Many pod Querying?



Ph.D. Symp. Paper



Contact